

Automatic DC-Offset Calibration for a Low-Cost Geophone Front End

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Abstract—This paper presents a foreground calibration method for a low-cost geophone analog front end in which stage-to-stage accumulation of amplifier and reference offsets can saturate intermediate nodes. The method sequentially measures and trims four cascaded stages – a programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP) – using existing on-chip conversion and filtering resources. Oscilloscope measurements on two independently assembled boards, with reported settings spanning the PGA's $\times 1$ – $\times 50$ gain range, show that calibration leaves the ADC-facing LP stage within 4% of the ADC's full-scale span in all eight reported board/gain combinations, down from 46% under the fixed-reference baseline. The PGA converges only partially in some conditions; a targeted test showed deadband-gated convergence rather than a hardware floor, while downstream correction preserved final-stage headroom. A four-stage run takes approximately 10 s typical (57.6 s worst case). Coherent multichannel surface-wave field records further demonstrate operation beyond the bench.

Index Terms—Geophone, analog front end, DC-offset calibration, mixed-signal systems, PSoC

I. INTRODUCTION

Multichannel surface-wave methods use distributed vibration measurements to estimate Rayleigh-wave dispersion and infer near-surface mechanical properties [1], [2]. Although their processing is well established, acquisition hardware remains a major source of cost and deployment complexity; compact wireless nodes based on inexpensive geophones and mixed-signal microcontrollers therefore enable denser, reconfigurable arrays [3]. Embedded interfaces already combine amplification, filtering, and programmable gain [4], but achieving low offset through higher-grade analog components raises the cost of every node. This work instead uses resources already present in the microcontroller to correct the operating point of a budget-constrained front end.

The system belongs to a low-cost, reconfigurable Internet-of-Things acquisition platform [5] based on SM-24 moving-coil geophones [6]. Its PSoC 5LP integrates programmable-gain and operational amplifiers, analog filters, voltage DACs, multiplexers, a delta-sigma ADC, a digital filter, DMA, and an Arm Cortex-M3 processor [7]. Time-sharing the ADC and digital filter between acquisition and calibration avoids a separate converter or processor, following digitally assisted analog design [8]. The tradeoff is that the internal analog blocks

cannot be independently optimized for offset, drift, and flicker noise, as also reported for a single-chip PSoC potentiostat [9].

The geophone conditioning circuit is derived from the cascade correction and subtraction topology of Ma *et al.* [10]; mapped to the PSoC, accumulated DC errors prevented sufficiently aggressive gain settings and became particularly severe at the SUM output.

A continuously active DC servo, evaluated as an initial solution, proved impractical (Section V); the proposed method instead performs a finite foreground calibration before measurement and disconnects the controller afterward.

On-chip self-calibration is established. PSoC-Stat injects five IDAC levels and uses the ADC response to estimate TIA/ADC gain and offset [9]; Infineon's AN68403 measures a grounded PGA/ADC chain under an internal VDAC and stores numerical corrections in EEPROM [11]. Physical trimming has also been demonstrated for a single amplifier through body-bias calibration [12]. Here, instead of correcting only the ADC transfer characteristic or one amplifier, the loop adjusts the physical reference of every stage in a cascade within the available DAC range and resolution.

The front end remains DC-coupled because the SM-24's usable band extends toward its mechanical corner: an input coupling capacitor would require a corner well below 1 Hz to avoid attenuating the targeted low-frequency content. AC blocking is therefore unsuitable, while substituting lower-offset external amplifiers would abandon the single-chip premise; per-stage trimming addresses offset without either compromise, unlike classical autozero/chopper techniques for offset and 1/f noise [13].

The contribution is threefold: 1) a per-stage analog calibration architecture using the existing PSoC multiplexer, ADC, digital filter, and four VDACS; 2) a foreground controller with deadband, anti-windup, slew limiting, stability detection, and one-LSB refinement that freezes its corrections before acquisition; and 3) validation on two independently assembled boards at settings spanning the PGA's $\times 1$ – $\times 50$ gain range, showing bounded offsets without per-board or per-gain manual retuning.

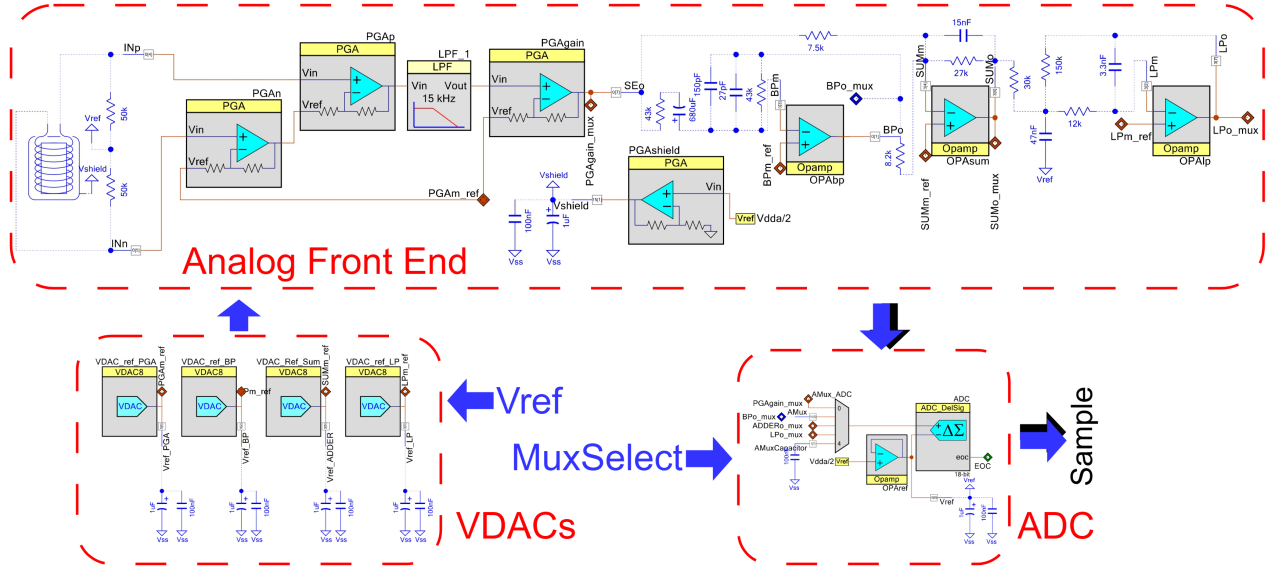


Fig. 1. Analog signal chain: four independent VDAC references feed the programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP) cascade, with the shared ADC/multiplexer measurement path at the output.

II. PLATFORM DESIGN

The PSoC conditions and digitizes the geophone output in acquisition mode. In calibration mode, its measurement path, digital-filter coefficients, and DAC-control firmware form a temporary mixed-signal loop without changing the sensor input chain.

In the cascade of Fig. 1, an offset introduced at any stage propagates through every downstream gain toward the ADC input. The calibration therefore exposes and trims each stage separately rather than correcting only the final ADC value.

An analog multiplexer connects one stage output at a time to the shared 18-bit delta-sigma ADC, which serves as the error sensor; software mutual exclusion prevents simultaneous connections. A switched capacitor behind the selected multiplexer MOSFET combines with its on-resistance for limited RC anti-aliasing and isolates the amplifier from a direct capacitive load.

For the i th stage, its slowly varying output can be represented as

$$y_i^{\text{DC}} = A_i y_{i-1}^{\text{DC}} + b_i + K_i u_i, \quad (1)$$

where A_i is the stage gain, b_i the amplifier and reference errors, K_i the signed transfer from the trim DAC to the measured node, and u_i the DAC code. Before per-stage calibration, the existing SUM reference was adjusted to recenter that node. This single-point correction left the remaining y_i^{DC} values uncontrolled: an earlier rail could not be recovered, offsets after SUM remained uncompensated, and the approach did not extrapolate to front ends without an explicit summing node.

The proposed architecture instead chooses each u_i in Eq. 1 independently so that every measured y_i^{DC} approaches its stage reference. In the differential geophone front end considered here, all four calibration targets correspond to zero ADC counts.

Eq. 1 is lower-triangular in stage index: y_i^{DC} depends on y_{i-1}^{DC} but not on downstream stages. Processing PGA, BP, SUM, and LP in signal-flow order therefore places y_{i-1}^{DC} at its target before stage i is trimmed, making a single forward pass sufficient (Section III).

III. SELF-CALIBRATING GEOPHONE PLATFORM

A. Digital Hardware and Calibration Sequence

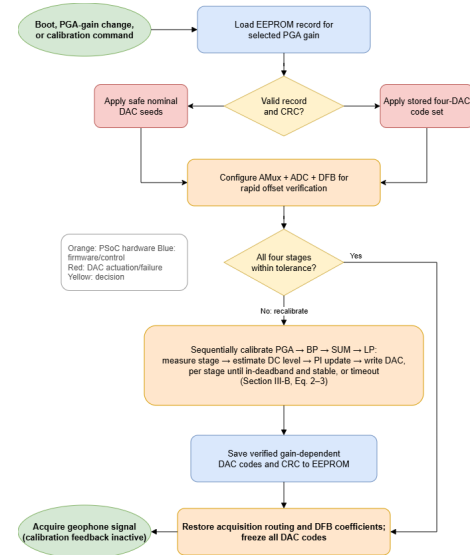


Fig. 2. Flowchart of the proposed foreground autocalibration algorithm, including stored-code verification, sequential per-stage correction, and transition to acquisition.

At boot, after a gain change, or on command, the firmware loads stored DAC codes and verifies the resulting offsets (Fig. 2). If all stages are within tolerance, it restores acquisition without a full run; otherwise it processes PGA, BP, SUM, and

LP sequentially, resetting the FIR history and settling after each multiplexer switch. A watchdog and per-stage sample limit bound the sequence. During acquisition the controller is inactive, the multiplexer selects LP, and the VDAC codes remain constant, so no time-varying feedback enters the acquired signal.

The shared digital filter implements the geophone response during acquisition and switches to an aggressive low-pass estimator during calibration (Fig. 3); its acquisition coefficients and path are then restored. Successful DAC codes are stored in EEPROM as one record per PGA gain setting containing the gain code, valid-stage mask, four DAC codes, version, and CRC. A failed run never overwrites a valid record.

B. Control Law and Calibration Parameters

For each stage, the filtered measurement $\hat{y}_i[n]$ produces the ADC-domain error $e_i[n] = r_i - \hat{y}_i[n]$, where $r_i = 0$ for the current differential geophone configuration. The error is first mapped to the 8-bit DAC domain:

$$\varepsilon_i[n] = \text{round}\left(\frac{e_i[n]V_{\text{ADC}}N_{\text{DAC}}}{N_{\text{ADC}}V_{\text{DAC}}}\right), \quad (2)$$

with $V_{\text{ADC}} = 5000$ mV, $N_{\text{ADC}} = 262144$, $V_{\text{DAC}} = 4080$ mV, and $N_{\text{DAC}} = 255$ in the implemented configuration.

The rounding in Eq. 2 makes $\varepsilon_i[n]$ an integer error in equivalent DAC codes. If the ideal operating point lies between two realizable codes, an acceptance band narrower than one output step can make the controller alternate indefinitely between them. The implemented deadband therefore scales with the DAC-to-node gain,

$$\Delta_i = \max(1, \lceil 1.2|K_i| \rceil), \quad (3) \quad \bar{\varepsilon}_i[n] = \begin{cases} 0, & |\varepsilon_i[n]| \leq \Delta_i, \\ \varepsilon_i[n], & \text{otherwise,} \end{cases} \quad (4)$$

where $\bar{\varepsilon}_i[n]$ is the error presented to the controller. The 1.2 factor provides a small margin over one realizable stage-output step, while the one-code floor preserves a finite acceptance region when $|K_i| < 1$.

The architecture requires only a measurable stage output and sufficient trim range; this prototype uses a positional PI law:

$$u_i^*[n] = u_{i,0} + s_i \left[\frac{k_{p,i}}{|K_i|} \bar{\varepsilon}_i[n] + \frac{k_{i,i}}{|K_i|} \sum_{q=0}^n \bar{\varepsilon}_i[q] \right], \quad (5)$$

Here $u_{i,0}$ is a feedforward code near the expected operating point, $s_i \in \{-1, +1\}$ is the injection polarity, and $|K_i|$ is the nominal stage movement per VDAC code in the scaled domain of Eq. 2. The fixed BP, SUM, and LP values are 2, 7.89, and 6, respectively; PGA K_i and s_i follow the active gain code. The feedforward codes were hand-set once on Board B at the power-up gain $\times 50$ and then used unchanged for Board A and all tested gains. With $k_{p,i} = 0$ for all stages, Eq. 5 reduces to the pure integrator parameterized by $k_{i,i}$ in Table I.

The command is clamped to the VDAC range and slew-limited to one code per sample, with conditional integration for anti-windup. Convergence requires a fixed number of samples in the same quantized bucket; a final adjacent-code trial is

TABLE I
IMPLEMENTED PER-STAGE CALIBRATION PARAMETERS. DEADBAND Δ_i IS GIVEN IN EQUIVALENT DAC CODES; THE DC ESTIMATOR RUNS AT THE ADC NATIVE SAMPLE RATE OF 2604 HZ.

Stage	K_i	Δ_i	$k_{i,i}$	Timeout (s)	Stage	K_i	Δ_i	$k_{i,i}$	Timeout (s)
PGA	gain-dependent	gain-dependent	3×10^{-4}	11.5	SUM	7.89	10	5×10^{-4}	11.5
BP	2	3	5×10^{-4}	17.3	LP	6	8	5×10^{-4}	17.3

retained only when it lowers the filtered error. At timeout, the best code found is held, the stage is flagged not converged, and the sequence advances.

Table I lists the implemented parameters. Settling uses 512 samples (0.20 s) for PGA/BP/SUM and 1024 (0.39 s) for LP; lock requires 512 or 1024 consecutive in-deadband samples, and refinement adds 1024–2048 samples. The listed timeouts are worst-case ceilings.

The 8-bit VDAC resolution meets this prototype's acceptance range. Applications requiring a smaller residual can trade added hardware for a finer step through a PSoC IDAC/resistor or a buffered topology [14], while retaining the same measure–estimate–correct sequence.

IV. EXPERIMENTAL VALIDATION

A. Prototype and Comparison Conditions

Two independently assembled SM-24 conditioning prototypes (Board A and Board B), each with four VDAC references, were tested under two conditions: *Default*, references hand-tuned once on Board B at gain $\times 50$ and reused unchanged; and *Proposed*, sequential four-stage calibration followed by frozen-reference acquisition. Default is therefore a fixed baseline calibrated for one board and operating point, not an uncalibrated circuit. The reported settings span the PGA's $\times 1$ – $\times 50$ range and test whether per-board and per-gain retuning is necessary; Table II reports the resulting per-stage offsets.

B. Design Target: Headroom, Not Zero

The loop targets headroom rather than exact zero. Its deadband, timeout, estimator, and gains balance residual error against convergence time within the available DAC, ADC, and FIR capabilities (Section III). The ADC-facing signal is centered on a 2.5 V reference, so its 0–5 V input span corresponds to a ± 2.5 V excursion about that operating point. Every measured node must remain in its linear, unsaturated range; Section V-A examines a case in which this criterion is met despite partial PGA convergence.

C. Measured Headroom Recovery Across the Gain Range

At each reported board/gain setting, a Tektronix TDS 2004B oscilloscope (4 channels, 60 MHz, 1 GS/s) measured the mean PGA, BP, SUM, and LP output over a 100 ms post-settling window under both conditions. Table II gives absolute DC offset as a percentage of this 5 V span, with $N=10$ independent runs per board/gain/stage cell. LP is the ADC-facing node; PGA, BP, and SUM expose upstream saturation or nonlinearity hidden by a final-output measurement alone. Board B $\times 1$ and $\times 4$ were omitted because the Default PGA offset was already inside the normal deadband, so the loop correctly converged

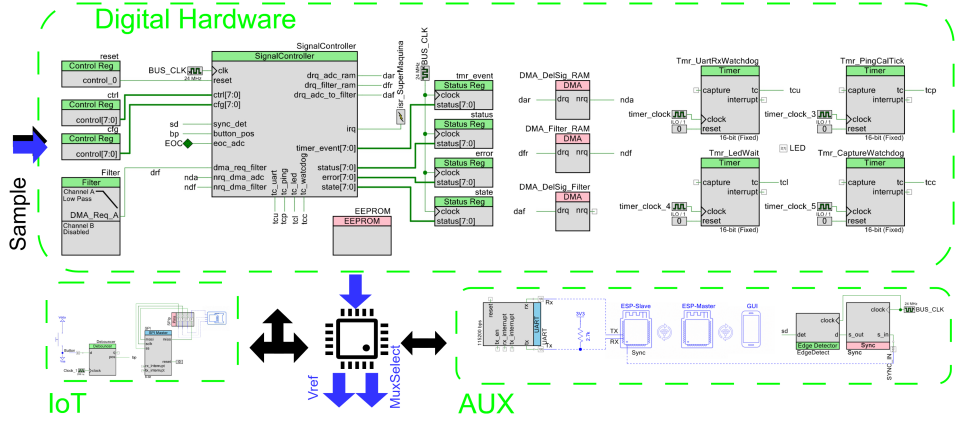


Fig. 3. Digital calibration hardware. The controller routes DMA transfers among the ADC, shared digital filter, and EEPROM and reconfigures the analog multiplexer. Black and blue arrows indicate acquisition and calibration paths, respectively.

TABLE II
RESIDUAL DC OFFSET AT EACH STAGE AS % OF THE ADC'S 0–5 V SPAN (± 2.5 V ABOUT THE 2.5 V REFERENCE), REPORTED AS DEFAULT / PROPOSED ($N=10$ PER CELL; LOWER IS BETTER).

Gain	Board A				Board B			
	PGA	BP	SUM	LP	PGA	BP	SUM	LP
$\times 1$	$3.5 \pm 0.2 / 3.4 \pm 0.3$	$0.8 \pm 0.2 / 1.2 \pm 0.2$	21.8 ± 1.1 / 0.8 ± 0.2	45.8 ± 3.0 / 3.2 ± 0.8	—	—	—	—
$\times 4$	$2.4 \pm 0.1 / 2.1 \pm 0.3$	$4.9 \pm 0.3 / 1.0 \pm 0.2$	$6.7 \pm 1.0 / 1.0 \pm 0.3$	42.6 ± 3.9 / 3.9 ± 0.8	—	—	—	—
$\times 16$	$2.4 \pm 0.1 / 1.9 \pm 0.2$	$3.8 \pm 0.9 / 1.5 \pm 1.5$	$4.9 \pm 1.5 / 1.4 \pm 1.5$	29.6 ± 6.5 / 2.3 ± 1.3	$2.1 \pm 0.1 / 1.7 \pm 0.1$	$0.66 \pm 0.02 / 0.66 \pm 0.02$	$4.5 \pm 0.8 / 0.6 \pm 0.5$	$3.0 \pm 0.1 / 0.7 \pm 0.1$
$\times 32$	$1.0 \pm 0.2 / 0.5 \pm 0.2$	$7.0 \pm 0.4 / 1.0 \pm 0.5$	$7.4 \pm 0.3 / 0.6 \pm 0.2$	12.4 ± 1.3 / 1.7 ± 0.6	$1.0 \pm 0.2 / 0.5 \pm 0.2$	$0.66 \pm 0.01 / 0.66 \pm 0.01$	$8.0 \pm 0.8 / 1.2 \pm 0.3$	$2.9 \pm 0.1 / 0.4 \pm 0.2$
$\times 50$	$1.2 \pm 0.3 / 1.0 \pm 0.5$	$1.0 \pm 0.7 / 0.9 \pm 0.5$	$0.4 \pm 0.2 / 0.8 \pm 0.6$	$4.1 \pm 1.9 / 1.2 \pm 0.4$	$2.5 \pm 0.1 / 1.8 \pm 0.3$	$0.65 \pm 0.01 / 0.65 \pm 0.02$	$0.9 \pm 0.6 / 0.9 \pm 0.6$	$2.9 \pm 0.1 / 0.7 \pm 0.2$

Red: Default $\geq 10\%$ FS. Orange: Proposed $\geq$ Default (Section V-A). Board B $\times 1/\times 4$ excluded: loop converged immediately without changing a DAC.

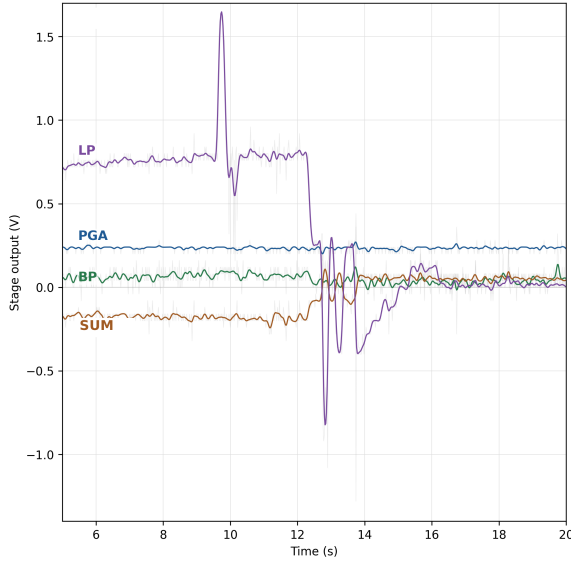


Fig. 4. Oscilloscope-derived PGA, BP, SUM, and LP outputs during a calibration run at PGA gain $\times 16$. Fixed references are active for $t \lesssim 9$ s; all stages converge by $t \approx 16$ s. The larger PGA residual is discussed in Section V-A.

without changing a DAC. Forcing action required manually narrowing the deadband and hence changing the test condition (Section V-A).

Across the eight reported board $\times$ gain cells, the LP mean offset spans 0.4–3.9%FS after calibration versus 2.9–45.8%FS under Default (Table II). Board A's Default offset decreases

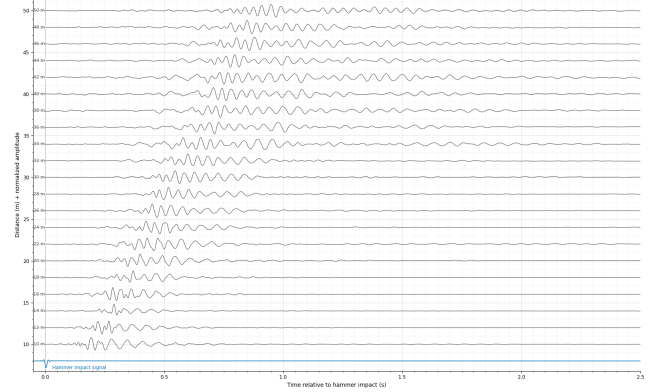


Fig. 5. Multichannel waterfall field record acquired with the calibrated front end.

with gain, whereas Board B remains closer to zero at its reported settings; Proposed brings both boards below 4%FS. Thus, within the tested boards and settings, calibration converts a board- and gain-dependent operating point into a bounded ADC-facing offset without manual retuning.

D. Field Use of the Calibrated Front End

Separate from the bench offset test, Fig. 5 shows a field-surface-wave waterfall over a 10–50 m line. First-arrival moveout is resolved at every offset using higher PGA settings at far offsets; this demonstrates field operation but is not a controlled Proposed-vs-fixed-reference comparison.

V. DISCUSSION

The method trades component precision for system-level measurement and digital assistance, unlike precision front ends characterized near the sensor's Johnson-noise limit [15]. Its benefit is not that the ADC output has zero mean – a software subtraction could do that – but that the DACs modify the physical operating points before large downstream gains apply, so the correction is agnostic to the error source, provided the stage is measurable and the DAC has sufficient authority and range.

The foreground method replaced a tested continuous DC servo: a low corner settled impractically slowly, a higher corner interfered with the signal, and each gain change restarted the transient, matching the corner-frequency/settling tradeoff reported for DC-servo compensation [16]. The proposed controller disconnects during acquisition, holding the DAC outputs constant and adding no time-varying feedback or servo poles. EEPROM avoids most full runs while the errors remain repeatable; if aging, temperature, or another condition moves a stored point outside tolerance, verification rejects it and recalibrates at boot, after a gain change, or on command (Section III). Remaining limitations are finite DAC resolution, no correction of noise or distortion, and unvalidated convergence under sustained vibration or thermal drift.

A. The PGA Stage: A Partial Result, Explained

Unlike BP, SUM, and LP, the PGA stage's post-calibration offset is not consistently much smaller than Default: the mean |offset| falls only modestly at low gain and by roughly half at best. The observed behavior is consistent with two firmware choices. First, the deadband Δ_i (Section III) is an acceptance margin, not a physical limit. On Board B at $\times 4$, the default PGA offset fell inside the normal deadband on every run attempted, and the controller declared convergence without moving the DAC. Tightening the deadband by hand caused the controller to act, directly confirming deadband-gated convergence in that condition. Second, the 128-sample FIR estimator trades residual resolution against convergence time; whether a longer estimator materially improves the remaining PGA offset was not tested. The deadband intervention rules out a fixed PGA hardware floor in the tested condition, and neither effect compromises the final-stage headroom target in Table II.

The two orange cells in Table II occur where the fixed-reference offset was already below 1%FS: BP at $\times 1$ and SUM at $\times 50$ on Board A. Their Proposed values remain at or below 1.2%FS. The controller is designed to enter an acceptance band, not to guarantee a lower sample mean in every repeated cell, so these small reversals do not contradict the headroom result.

Fig. 4 shows the sequential PGA $\rightarrow$ BP $\rightarrow$ SUM $\rightarrow$ LP correction: each measured output steps when addressed and then holds. The four-stage sequence takes approximately 10 s typical; the Table I timeouts sum to 57.6 s only if every stage times out, which was not observed.

The two boards have markedly different Default operating points, yet both satisfy the headroom target at their reported settings. Feedforward values were tuned only on Board B

at $\times 50$; Board A received no manual adjustment and still converged across its tested gains. This supports reuse of one nominal initialization in a dense network, with the loop absorbing unit- and gain-dependent variation. The $N=10$ runs quantify within-cell repeatability, but two boards cannot establish population-level scaling; more units and cross-temperature trials are required.

VI. CONCLUSION

A per-stage foreground calibration method was implemented in a low-cost PSoc geophone front end using four VDACS, the existing analog multiplexer, acquisition ADC, and digital filter. It trims the physical operating point of each stage, freezes the resulting codes during acquisition, and verifies stored gain-dependent corrections before reuse.

Across eight reported board/gain cells on two boards, collectively spanning $\times 1$ – $\times 50$, the ADC-facing LP mean offset was 0.4–3.9%FS after calibration versus 2.9–45.8%FS under the fixed-reference baseline, without per-board or per-gain retuning. A targeted test attributed partial PGA convergence to deadband gating in that condition, while final-stage headroom remained below 4%FS. Calibration takes approximately 10 s typical (57.6 s worst case). Field records demonstrate operation over a 10–50 m line; population scaling, temperature dependence, VDACS-reference noise, and a controlled field comparison remain to be measured.

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